

ELITE MATHEMATICS

Pearson Edexcel International GCSE

Linear 4MA1

Adaptive Classified

OL-T27 · Simultaneous Equations

Questions

Student book · questions with writing space

■ 26 pages

FAMILY MAP

Simultaneous Equations

F01	Solve a linear-quadratic system	11 Q 57 marks
F02	Eliminate between two linear equations	6 Q 18 marks
F03	Interpret a system graphically	6 Q 23 marks
F04	Substitute or model a linear system	1 Q 5 marks

22 Solve the simultaneous equations

$$\begin{aligned}x^2 + 3y + y^2 &= 7 \\ y &= x + 2\end{aligned}$$

Show clear algebraic working.

(Total for Question 22 is 5 marks)

$$\begin{array}{rcl} 3x + 5y & = & 8 \\ 4x + y & = & -3.5 \end{array}$$

$x =$

$y =$

Question 16

4MA1-2H Q10

MODULAR UNIT 2

3 marks

10 Solve the simultaneous equations

$$6x + 4y = 1$$

$$3x + 5y = 8$$

Show clear algebraic working.

 $x = \dots\dots\dots$ $y = \dots\dots\dots$

Question 25

4MA1-2H Q22

MODULAR UNIT 2

5 marks

22 The straight line **L** has equation $x + y = 5$

The curve **C** has equation $2x^2 + 3y^2 = 210$

Find the coordinates of the points where **L** and **C** intersect.

Show clear algebraic working.

(.....,) (.....,)

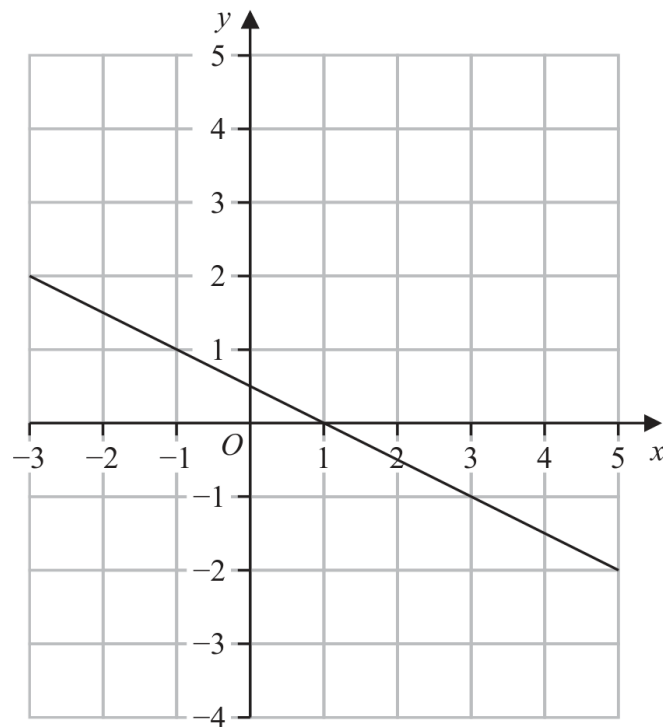
Question 29

4MA1-2H Q14 whole

MODULAR UNIT 2

3 marks

- 14** Here is the graph of the equation $2y + x = 1$ drawn on a grid.



By drawing another straight line on the grid, solve the simultaneous equations

$$y - x - 2 = 0$$

$$2y + x = 1$$

$x =$

Question 34

4MA1-2H Q21 whole

MODULAR UNIT 2

5 marks

- 21 The line with equation $x + 2y = 5$ intersects the curve with equation $x^2 + 3y^2 = 13$ at the points A and B

Find the coordinates of A and the coordinates of B
Show clear algebraic working.

(..... ,)

(..... ,)

4MA1-1H Q9

3 marks

$$7x + 3y = 8$$

$x =$

$y =$

Question 28

4MA1-2H Q21

MODULAR UNIT 2

5 marks

21 Solve the simultaneous equations

$$\begin{aligned}2x^2 + 3y^2 &= 11 \\ x &= 3y - 1\end{aligned}$$

Show clear algebraic working.

4MA1-2H Q22 M01

5 marks

$$x + 2 = y$$

.....

Question 24

4MA1-2H Q10 Q10

MODULAR UNIT 2

3 marks

10 Solve the simultaneous equations

$$7x + 3y = 3$$

$$3x - y = 7$$

Show clear algebraic working.

 $x = \dots\dots\dots$ $y = \dots\dots\dots$

Question 12

4MA1-1H Q19

MODULAR UNIT 2

5 marks

19 Solve the simultaneous equations

$$\begin{aligned} 3x^2 + y^2 - xy &= 5 \\ y &= 2x - 3 \end{aligned}$$

Show clear algebraic working.

.....

Question 24

4MA1-2H Q10

MODULAR UNIT 2

3 marks

10 Solve the simultaneous equations

$$3x + 5y = 3.1$$

$$6x + 3y = 3.75$$

Show clear algebraic working.

 $x =$ $y =$

Find the coordinates of A and B .
Show clear algebraic working.

(.....,)

$$x + 2y - 1 = 0$$

Show clear algebraic working.

$$\begin{array}{rcl} 5a + 2c & = & 10 \\ 2a - 4c & = & 7 \end{array}$$

$a =$

$$\begin{aligned} y &= 3 - 2x \\ x^2 + y^2 &= 18 \end{aligned}$$

Show clear algebraic working.

- Work out how much Michael pays for his 10 apples and 10 pears.
Show your working clearly.

- 19 The straight line **L** has equation $x - y = 3$
The curve **C** has equation $3x^2 - y^2 + xy = 9$

L and **C** intersect at the points P and Q .

Find the coordinates of the midpoint of PQ .
Show clear algebraic working.

(.....,)

12 Solve the simultaneous equations

$$7x - 2y = 34$$

$$3x + 5y = -3$$

Show clear algebraic working.

$$x = \dots\dots\dots$$

$$y = \dots\dots\dots$$

(Total for Question 12 is 4 marks)

$$\begin{aligned} 3xy - y^2 &= 8 \\ x - 2y &= 1 \end{aligned}$$

Show clear algebraic working.

$$\begin{array}{r} 5x + 4y = -2 \\ 2x - y = 4.4 \end{array}$$

$x =$

$y =$

(Total for Question 7 is 3 marks)

$$\begin{aligned} y &= 7 - 2x \\ x^2 + y^2 &= 34 \end{aligned}$$

Show clear algebraic working.

Work out the coordinates of B
Show your working clearly.

(..... ,)

(Total for Question 28 is 6 marks)

$$\begin{array}{r} 5x + y = 11 \\ 3x - y = 9 \end{array}$$

$x =$

$y =$

(Total for Question 2 is 3 marks)